

1. I can explain the purpose of stomata and how they open and close : written
2. I can discuss one adaptation that plants have to reduce water loss through stomata: written
3. I can label a cross section of a leaf and i know the major function of each part: written
4. I can draw and explain a simple representation of a carbon cycle and label it with the appropriate vocabulary (photosynthesis, respiration, decomposition, combustion, diffusion). : written
5. I can write a balanced chemical equation for photosynthesis and respiration : written
6. Given a chemical equation, i can identify which substances are the reactants and which substances are the products : multiple choice
7. I can describe one way that scientists use trees to study climate history(stomata, pollen cores, leaf edges) : written
8. I can describe several positive feedback loops related to climate change.

Investigating leaf stomata

Introduction

During photosynthesis, plants use carbon dioxide, water, and light to make sugar. In order to accomplish this, plants must absorb carbon dioxide and release oxygen. Gas exchange between the plant and atmosphere occurs in the leaf of a plant through tiny pores called stomata. Besides being a pathway for the uptake of carbon dioxide, these stomata are also a source of water loss through evaporation. This creates a dilemma for many plants. How can plants maximize the uptake of carbon dioxide and minimize water loss? Among other methods, plants are able to regulate the opening and closing of the stomata.

The stomata pores are surrounded on both sides by jelly bean shaped cells called guard cells. When guard cells swell with water, a gap forms between the cells and the stoma opens. Guard cells also close when dehydrated when water is removed from the guard cells, the cells shrink and the stoma closes. In this way, the plant can regulate the uptake of carbon dioxide and the loss of water.

Most stomata are on the lower epidermis of the leaves on most plants. Some species, like corn, have stomata on both the upper and lower epidermis and some species, like water lilies, only have stomata on the upper epidermis. Many plants leave their stomata open most of the time so they can acquire CO₂ for photosynthesis. Some plants, such as corn and crabgrass, close their stomata during the hottest part of the day; this reduces water loss. Plants such as pineapple and cacti that live in extremely hot, dry environments only open their stomata at night. This reduces water loss, but because they can only take in CO₂ at night, they grow slowly. The number and location of stomata can reveal much about the environment in which a plant lives.

Carbon cycle reading

What is carbon?

Carbon is the 6th element on the periodic table. In its simplest form. It is an atom with 6 protons, 6 neutrons, and 6 electrons. The protons and neutrons are found in the nucleus of the atom, and the electrons are found in shells around the nucleus. The first shell holds 2 electrons, and the second shell holds 5. This is a critical key to carbon's importance to life. To become stable, carbon needs to fill its shell with 4 more electrons, so it bonds with 4 other atoms simultaneously. This allows it to build a vast array of stable molecules from carbohydrates, to proteins, to fats and DNA. trees, humans, fish, and all other life on earth are built of carbon. After removing the hydrogen and oxygen from a living thing, the next most common element you'll find is carbon.

Carbon in the atmosphere

Carbon is also floating in the air around us in two main forms: methane and carbon dioxide. Both of these gasses are good insulators, which means they trap heat and contribute to the greenhouse effect. The greenhouse effect is the accumulation of heat in earth's atmosphere as the sun's warmth is not able to escape as quickly as it collects.

Carbon dioxide is the waste product of cellular respiration, the process all living things use to release energy from food. In other words, as all life releases the energy they need to live, carbon dioxide is created as waste. With all life exhaling carbon dioxide, our earth could heat up very quickly. Fortunately, this doesn't happen. Plants are the hero of this story. Plants conveniently need carbon dioxide to pair with water and sunlight during photosynthesis as they make sugar. The combination of all the plants on earth doing photosynthesis keeps the carbon dioxide under control and our earth stays cool as a result.

Carbon in the earth

When living things die, their bodies decompose, and carbon that made them is returned to soil. If lots and lots of living things die in the same area, massive collections of carbon could form into what we call fossil fuels. Crude oil, coal, and gas are fossil fuels that are mined and are the carbon and hydrogen left over from organisms that died millions of years ago. They decomposed until only carbon and hydrogen remained. It just so happens that carbon hydrogen bonds are highly flammable and make a wonderful fuel. You don't have to wait a million years to prove it either. Even a tree that has been dead for a couple of days will burn. It's not the water in the trees that's burning, so what is it? It's the carbon hydrogen bonds! Crude oil, coal, and gas are simply very highly concentrated versions of that log. Fossil fuels are extremely useful for keeping our fast paced society on the go, powering cars, planes, and electrical plants, but there's a catch.

In the water cycle, when water evaporates, does it disappear? No, it simply changes form. The carbon cycle is the same. When we burn oil, coal, and gas that has been trapped underground for millions of years, where does that carbon go? It's released in the air as carbon dioxide. These industrial emissions contribute to the warming of our planet through the greenhouse effect.

Carbon in the ocean

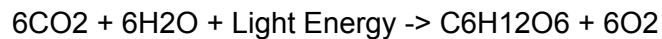
Just as oxygen can be dissolved in water, so can carbon through a process called diffusion, as a matter of fact the oceans are the world's largest "carbon sink" or place where carbon dioxide is removed from the air and stored. In fact, the oceans absorb about half of the carbon dioxide

that is emitted into the atmosphere. Some of the carbon in the ocean is incorporated into the bodies of living things, especially in shells. Unfortunately, as the oceans absorb more carbon, they become more acidic; this weakens the shells.

As you can see there is a delicate balance to the carbon cycle, and the recent rapid release of carbon dioxide into the atmosphere has disturbed that balance. While fossil fuels have been an amazing source of energy for our civilization, it may be time to find other sources of fuel and find ways of sequestering the excess carbon dioxide.

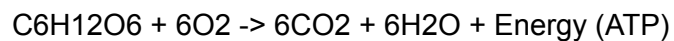
Equations:

Photosynthesis:



In words: carbon dioxide + water + light energy = glucose + oxygen

Respiration:



In words: glucose + oxygen = carbon dioxide + water + energy

Reactants vs products rules

- Reactants : starting materials
- Products: resulting substances
- Photosynthesis: reactants = $6\text{CO}_2 + 6\text{H}_2\text{O}$ | Products = $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- respiration : reactants = $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ | Products = $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$

Scientists use pollen cores to reconstruct past climate history by analyzing the layer by layer accumulation of microscopic pollen grains preserved in lake beds, peat bogs, and ocean sediments. Because pollen grains are coated in sporopollenin one of the most chemically resistant organic polymers known, their durable outer shells can survive and be intact for thousands or even millions of years without decaying, paleoclimatologists extract vertical sediment cylinders using specialized hollow drills, taking advantage of the law of superposition, which dictates that deeper sediment layers represent older time periods. By applying

radiocarbon dating to organic materials within specific layers, scientists establish an accurate timeline for the core, samples from each layer are examined under microscopic magnification, where researchers classify and count pollen grains based on their unique morphology, surface texture, and aperture shapes. Because every plant species thrives only within specific environmental niches defined by distinct temperature ranges, precipitation levels, and soil conditions, changes in pollen composition directly reflect changes in the surrounding terrestrial ecosystem over time. For instance, a sediment layer dominated by spruce or fir pollen signals a cold, subarctic climate, where a transition in upper layers to oak, hickory, or grass pollen demonstrates a shift towards a warmer, drier environment. By tracking these shifting vegetation patterns, scientists can build precise regional climate models that map ancient temperature changes, moisture availability, and forest composition through deep geological time.

Feedback loops:

1 ocean release of carbon dioxide

The ocean absorbs about half of the carbon dioxide in the atmosphere acting as a carbon sink - a part of the environment that stores carbon. The upper layer of the ocean absorbs CO₂. The cooler the water temperature, the more CO₂ the water can hold. So as the earth gets warmer, the amount of carbon dioxide the ocean can hold is expected to decrease, and the CO₂ stored in the ocean is expected to be released into the atmosphere, which could, in turn, make the earth warmer, so then the ocean will release more CO₂, so then it will be warmer, this is an example of a positive feedback loop because the problem keeps adding to itself. In other words, the effect of the loop amplifies the input.

Best study methods for this paper:

1. Blurting method (best for written paragraphs)
 - a. How to do it: take a blank piece of paper or open a document. And set a timer for 3 minutes and write out everything you remember about that information.
 - b. Check information
 - c. Repeat.
2. Active recall / flash cards(best for equations & definitions)
3. Spaced practice schedule:

How to do it: run through all the information for 15 - 20 minutes, and wait 30 minutes -2 hours and do it again, until the day of the test.